

Travis Askham

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Education

2016	Ph.D., Mathematics. New York University
2010	M.A., Mathematics. University of California Los Angeles
2010	B.Sc., Applied Mathematics. University of California Los Angeles

Professional Appointments

2026–present	Associate Professor, Department of Mathematical Sciences, New Jersey Institute of Technology
2019–2026	Assistant Professor, Department of Mathematical Sciences, New Jersey Institute of Technology
2018–2019	Visiting Scientist, Plasma Science and Fusion Center, Massachusetts Institute of Technology
2016–2018	Research Associate, Department of Applied Mathematics, University of Washington
2019–present	Visiting Scholar, Center for Computational Mathematics, Flatiron Institute

Publications

Journal Articles & Thesis

- [1] Peter Nekrasov, Zhaosen Su, **Travis Askham**, Jeremy G. Hoskins, Boundary integral formulations for flexural wave scattering in thin plates. *Journal of Computational Physics*, 542:114355, December 2025.
- [2] **Travis Askham**, Carlos Borges, Jeremy Hoskins, Manas Rachh, Random walks in frequency and the reconstruction of obstacles with cavities from multi-frequency data. *Journal of Scientific Computing*, 98(1):15, 2024.
- [3] **Travis Askham**, Carlos Borges, Reconstructing the shape and material parameters of dissipative obstacles using an impedance model. *Inverse Problems*, 40(9):095004, 2024.
- [4] Ruqi Pei, **Travis Askham**, Leslie Greengard, Shidong Jiang, A fast method for imposing periodic boundary conditions on arbitrarily-shaped lattices in two dimensions. *Journal of Computational Physics*, 474:111792, 2023.
- [5] **Travis Askham**, Peng Zheng, Aleksandr Aravkin, J Nathan Kutz, Robust and scalable methods for the dynamic mode decomposition. *SIAM Journal on Applied Dynamical Systems*, 21(1):60–79, 2022.
- [6] Ludvig af Klinteberg, **Travis Askham**, Mary Catherine Kropinski, A fast integral equation method for the two-dimensional Navier–Stokes equations. *Journal of Computational Physics*, 409:109353, May 2020.

- [7] **Travis Askham**, Manas Rachh, A boundary integral equation approach to computing eigenvalues of the Stokes operator. *Advances in Computational Mathematics*, 46(2):20, 2020.
- [8] Emily Clark, **Travis Askham**, Steven L Brunton, J Nathan Kutz, Greedy sensor placement with cost constraints. *IEEE Sensors Journal*, 19(7):2642–2656, 2019.
- [9] Niall M Mangan, **Travis Askham**, Steven L Brunton, J Nathan Kutz, Joshua L Proctor, Model selection for hybrid dynamical systems via sparse regression. *Proceedings of the Royal Society A*, 475(2223):20180534, 2019.
- [10] Peng Zheng, **Travis Askham**, Steven L Brunton, J Nathan Kutz, Aleksandr Y Aravkin, A unified framework for sparse relaxed regularized regression: SR3. *IEEE Access*, 7:1404–1423, 2019.
- [11] **Travis Askham**, A stabilized separation of variables method for the modified biharmonic equation. *Journal of Scientific Computing*, 76(3):1674–1697, 2018.
- [12] **Travis Askham**, J Nathan Kutz, Variable projection methods for an optimized dynamic mode decomposition. *SIAM Journal on Applied Dynamical Systems*, 17(1):380–416, 2018.
- [13] Manas Rachh, **Travis Askham**, Integral equation formulation of the biharmonic Dirichlet problem. *Journal of Scientific Computing*, 75(2):762–781, 2018.
- [14] Chang Sun, **Travis Askham**, J Nathan Kutz, Stability and dynamics of microring combs: elliptic function solutions of the Lugiato–Lefever equation. *Journal of the Optical Society of America B*, 35(6):1341–1353, 2018.
- [15] **Travis Askham**, Antoine J Cerfon, An adaptive fast multipole accelerated Poisson solver for complex geometries. *Journal of Computational Physics*, 344:1–22, January 2017.
- [16] **Travis Askham**, Leslie Greengard, Norm-preserving discretization of integral equations for elliptic PDEs with internal layers I: the one-dimensional case. *SIAM Review*, 56(4):625–641, 2014.
- [17] **Travis Askham**, Integral-equation methods for inhomogeneous elliptic partial differential equations in complex geometry. Ph.D. thesis, New York University, 2016.

Preprints

- [1] Jacob Linden, **Travis Askham**, Jeremy Hoskins, A boundary integral formulation of an acoustic boundary layer model in 2D. arXiv:2601.17283, January 2026. [link]
- [2] **Travis Askham**, Tristan Goodwill, Jeremy G Hoskins, Peter Nekrasov, Manas Rachh, Surface layers and linearized water waves: a boundary integral equation framework. arXiv:2510.22748, October 2025. [link]
- [3] **Travis Askham**, Jeremy G Hoskins, Peter Nekrasov, Manas Rachh, Integral equations for flexural-gravity waves: analysis and numerical methods. arXiv:2501.00887, January 2025. Under review at Pure and Applied Analysis. [link]

Honors & Awards

2016	Wilhelm Magnus Memorial Prize, Courant Institute of Mathematical Sciences, New York University
2010	Daus Award in Mathematics, University of California Los Angeles

Grants & Fellowships

Grants

2025–2027 PI: Travis Askham. American Chemical Society (ACS), Petroleum Research Fund (PRF) New Directions Grant. Numerical methods and simulation for electrocoalescence of 3D drops. With Michael Siegel, NJIT.

Fellowships

2015 Dean's Dissertation Fellowship, New York University
2010–2015 Henry M. MacCracken Fellowship, New York University

Teaching Experience

New Jersey Institute of Technology

- Fast Numerical Algorithms (Math 767)
- Introduction to Functions of a Complex Variable (Math)
- Introduction to Partial Differential Equations (Math)
- Linear Algebra (Math, Math 337)
- Numerical Linear Algebra (Math 707)
- ST: Numerical Linear Algebra (Math)
- Vector Analysis (Math)

University of Washington

- Scientific Computing (AMATH)

Research Experience

2016–2018 Research Associate, Department of Applied Mathematics, University of Washington. Principal Investigator: J. Nathan Kutz. Project: Sparse measurements and optimal sensor placement for classification and state estimation of complex systems

2012–2015 Research Assistant, Courant Institute of Mathematical Sciences, New York University. Principal Investigator: Leslie Greengard. Project: Novel methods for electromagnetic simulation and design

Student Supervision

2026 (exp.) R. Connor Greene, Ph.D., NJIT. co-advisor with Michael Siegel, DMS

2023 Connor Robertson, Ph.D., NJIT. co-advisor with Anand Oza, DMS

2022 Ruqi Pei, Ph.D., NJIT. co-advisor with Shidong Jiang, DMS

Spring 2025–present David Pritchard, undergraduate, NJIT. Research advisor: analyzing mixed boundary value problems using the techniques of Serkh–Rokhlin (2016)

Spring 2025–present Shivam Bhavsar, undergraduate, NJIT. Research advisor: developing quadrature rules for nearly singular integrals

Fall 2022–Spring 2023 Arij Qureshi, undergraduate, NJIT. Research advisor: eigenvalues of the Stokes operator (and others) on domains with corners

Fall 2022–Spring 2023 Raaid Kabir, undergraduate, NJIT. Research advisor: eigenvalues of the Stokes operator (and others) on domains with corners

Invited Speaking

Seminars & Colloquia

- 2026 Integral formulations and numerical methods for a class of half-space interface problems. Purdue University. West Lafayette, IN, USA
- 2025 Integral equation methods and analysis for half-space surface wave problems. Center for Computational Mathematics Seminar, Flatiron Institute. New York, NY, USA
- 2025 Integral equation methods for flexural gravity waves (or, a Song of Ice and Water). Seminar of the Committee on Computational and Applied Mathematics, University of Chicago. Chicago, IL, USA
- 2023 Inverse obstacle scattering with trapping and dissipative media. Yale Applied Mathematics Seminar. New Haven, CT, USA
- 2021 A periodic fast multipole method. University of Central Florida Mathematics Colloquium. Online
- 2021 chunkIE: a MATLAB integral equation library. University of Massachusetts Lowell / University of North Carolina Computational Mathematics Seminar. Online
- 2018 A Fredholm operator approach to clamped plate problems. Simon Fraser University. Burnaby, BC, Canada
- 2018 Tailored low-rank matrix approximation: two stories. New Jersey Institute of Technology. Newark, NJ, USA

Conference Activity

Presentations

- 2025 Talk. Integral formulations and numerical methods for a class of half-space interface problems. Institut Mittag-Leffler Program on Interfaces and Unfitted Discretization Methods. Djursholm, Sweden
- 2025 Talk. Boundary integral methods for flexural wave scattering. SIAM Conference on Computational Science and Engineering (CSE25). Fort Worth, TX, USA
- 2023 Talk. Representing mixed boundary conditions and material interfaces in the chunkIE software. SIAM New York–New Jersey–Pennsylvania (SIAM-NNP) Regional Conference. NJIT, Newark, NJ, USA
- 2023 Talk. Exploring inverse obstacle scattering with an impedance model. International Congress on Industrial and Applied Mathematics (ICIAM23). Waseda University, Tokyo, Japan
- 2023 Talk. Function extension along normals. SIAM Conference on Computational Science and Engineering (CSE23). Amsterdam, the Netherlands
- 2022 Talk. A periodic fast multipole method. SIAM Conference on Imaging Science (IS22). Online
- 2021 Talk. A periodic fast multipole method. SIAM Annual Meeting (AN21). Online

- 2021 Talk. Fast multipole methods for continuous charge densities. SIAM Conference on Computational Science and Engineering (CSE21). Online
- 2020 Talk. Fast multipole methods for continuous charge densities. Canadian Mathematical Society (CMS) Winter Meeting. Online
- 2020 Talk. Fast multipole methods for continuous charge densities. SIAM Conference on Imaging Science (IS20). Online
- 2019 Talk. Fredholm determinants: a robust approach to computing Stokes eigenvalues. Canadian Applied and Industrial Mathematics Society (CAIMS) Annual Meeting. Whistler, BC, Canada
- 2019 Talk. Fredholm determinants: a robust approach to computing Stokes eigenvalues. SIAM Conference on Computational Science and Engineering (CSE19). Spokane, WA, USA
- 2019 Talk. Fredholm determinants: a robust approach to computing Stokes eigenvalues. International Congress on Industrial and Applied Mathematics (ICIAM19). Valencia, Spain
- 2018 Talk. A stable, kernel dependent FMM for fluid flow. ICOSAHOM. London, UK
- 2018 Talk. Robust and scalable methods for the dynamic mode decomposition. SIAM Conference on Uncertainty Quantification (UQ18). Garden Grove, CA, USA
- 2018 Talk. Adaptive grids for embedded integral equation based solvers. ICERM Workshop on Point Configurations. Providence, RI, USA
- 2017 Talk. Robust and scalable methods for the dynamic mode decomposition. SIAM Pacific Northwest Regional Meeting. Corvallis, OR, USA
- 2017 Talk. A stabilized FMM for fluid flow. BIRS-CMO Workshop on Creeping Flows. Oaxaca, Mexico
- 2017 Talk. Variable projection for generalizing the dynamic mode decomposition. SIAM Conference on Computational Science and Engineering (CSE17). Atlanta, GA, USA
- 2017 Talk. An algorithm for the DMD with unevenly spaced time samples. BIRS Workshop on Data-Driven Methods. Banff, AB, Canada
- 2016 Talk. Integral-equation methods for inhomogeneous elliptic PDEs (and applications). SIAM Annual Meeting. Boston, MA, USA
- 2013 Talk. On the discretization of integral equations for elliptic PDEs with internal layers. Mid-Atlantic Numerical Analysis Day, Temple University. Philadelphia, PA, USA
- 2014 Poster. Volume integrals in complex geometry: a case study of Poisson's equation. CBMS-NSF Conference: Fast Direct Solvers for Elliptic PDEs, Dartmouth College. Hanover, NH, USA
- 2013 Poster. On the discretization of integral equations for divergence-form PDEs with internal layers. Integral Equation Methods: Fast Algorithms and Applications (BIRS Workshop). Banff, AB, Canada

Organization

- 2024 Workshop. Computational Tools for PDEs with Complicated Geometries and Interfaces. Flatiron Institute, New York, NY, USA. June 10–14, 2024
- 2023 Minisymposium. MS31: Fast algorithms: community software and applications. SIAM NNP 2023 (with Manas Rachh, Flatiron Institute)

2023	Minisymposium. Integral equation and high frequency methods. FACM 2023 (with Yassine Boubendir, NJIT)
2023	Minisymposium. MS1: Fast algorithms for variable coefficient, time-dependent, and inhomogeneous problems. SIAM CSE 2023 (with Abinand Gopal, Yale)
2018	Minisymposium. High-order integral equation methods in fluid dynamics. ICOSAHOM 2018
2018	Minisymposium. Data-driven discovery for dynamical systems. SIAM UQ 2018
2017	Minisymposium. Data-driven characterization, control, and uncertainty quantification of dynamical systems. SIAM CSE 2017
2026	Hackathon. chunkIEfest2026a. Flatiron Institute, New York, NY. May 16–17, 2026
2025	Hackathon. chunkIEfest2025b. Flatiron Institute, New York, NY. October 27–29, 2025
2025	Hackathon. chunkIEfest2025a. NJIT, Newark, NJ and the Flatiron Institute, New York, NY. April 28–30, 2025
2024	Hackathon. chunkIEfest2024b. Meyer Sound LLC, Berkeley, CA. October 16–17, 2024
2024	Hackathon. chunkIEfest2024a. National Institute for Theory and Mathematics in Biology, Chicago, IL. April 1–2, 2024
2023	Hackathon. chunkIEfest2023. NJIT, Newark, NJ. October 31, 2023
2022	Hackathon. chunkIEfest2022b. NJIT, Newark, NJ and the Flatiron Institute, New York, NY. December 6–8, 2022

Service to the Profession

Referee

- *Advances in Computational Mathematics*
- *Boletín de la Sociedad Matemática Mexicana*
- *Communications on Pure and Applied Mathematics*
- *IEEE Transactions on Automatic Control*
- *Journal of Computational Physics*
- *Journal of Engineering Mathematics*
- *Journal of Scientific Computing*
- *Journal of Sound and Vibration*
- *Nonlinear Dynamics*
- *Proceedings of the Royal Society A*
- *SIAM Journal on Applied Dynamical Systems*
- *SIAM Journal on Imaging Sciences*
- *SIAM Journal on Numerical Analysis*
- *SIAM Journal on Scientific Computing*
- *SoftwareX*

Membership

- Society for Industrial and Applied Mathematics (SIAM) (since 2011)

Software

Maintainer

chunkIE	Pedagogical and prototyping code for solving integral equations on curves. Chunk-based discretization with adaptive refinement and high-order quadrature. MATLAB toolbox for boundary integral equations on curves, available under the 3-clause BSD license. https://github.com/fastalgorithms/chunkie
optdmd	Computing the optimized dynamic mode decomposition. MATLAB package, available under the MIT license. https://github.com/duqbo/optdmd
RobustDMD.jl	Fitting exponential functions to data with robust penalties. Julia package, available under the MIT license. https://github.com/UW-AMO/RobustDMD.jl
mbh2dfmm	Stable fast sums of the modified biharmonic Green's function and modified Stokes kernels. Fortran package, available under the modified FreeBSD license. https://github.com/duqbo/mbh2dfmm

Contributor

FMM3D	Flatiron Institute fast multipole libraries, for fast particle sums. Contributed new kernels and Julia wrappers. Fortran library with MATLAB and Python interfaces, available under the 3-clause BSD license. https://github.com/flatironinstitute/FMM3D
fmm3dbie	Integral equation methods in three dimensions. Contributed Stokes codes. Fortran library with MATLAB and Python interfaces, available under the 3-clause BSD license. https://gitlab.com/fastalgorithms/fmm3dbie
fmm2d	Fast particle sums in two dimensions. Contributed new kernels. Fortran library with high-level language wrappers, available under the 3-clause BSD license. https://github.com/flatironinstitute/fmm2d
inverse-obstacle-scattering2d	Inverse obstacle scattering in two dimensions. Contributed new options for the optimization routines and methods for impedance models. MATLAB package. https://github.com/flatironinstitute/inverse-obstacle-scattering2d

Skills

Mastery	Fortran (77–95), MATLAB
Proficiency	C99/C++, LaTeX, Python, Julia
Familiarity	OpenMP, OpenCL (in C99), PHP, HTML
Languages	English (native)

Biographical

Born	1987. Walnut Creek, CA, USA
Citizenship	United States